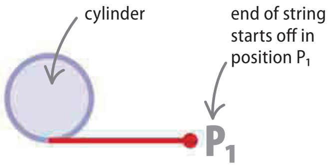
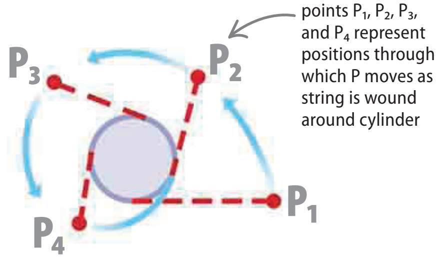
LOOKING CLOSER

Spiral locus

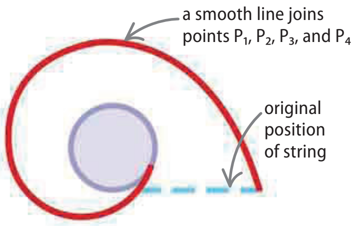
Loci can follow more complex paths. The example below follows the path of a piece of string that is wound around a cylinder, creating a spiral locus.



The string starts off lying flat, with point P_1 the position of the end of the string.



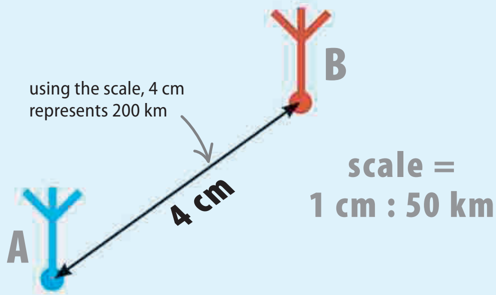
As the string is wound around the cylinder, the end of the string moves closer to the surface of the cylinder.



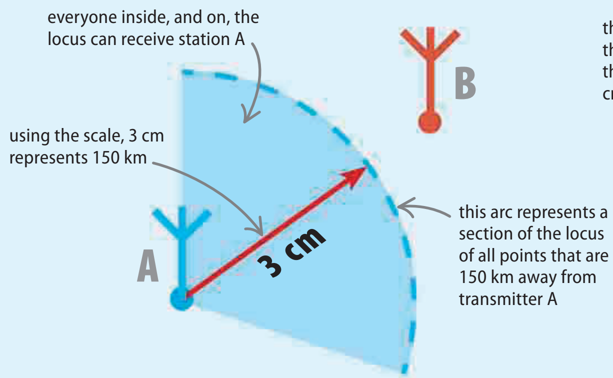
When the path of point P is plotted, it forms a spiral locus.

Using loci

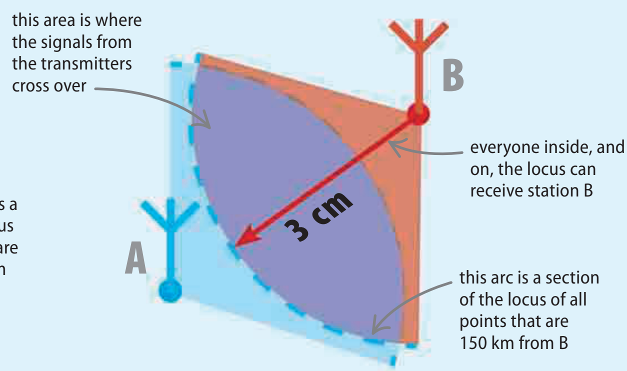
Loci can be used to solve difficult problems. Suppose two radio stations, A and B, share the same frequency, but are 200 km apart. The range of their transmitters is 150 km. The area where the ranges of the two transmitters overlap, or interference, can be found by showing the locus of each transmitter and using a scale drawing (see pp.106–107).



To find the area of interference, first choose a scale, then draw the reach of each transmitter. An appropriate scale for this example is 1 cm : 50 km.



Construct the reception area for radio station A. Draw the locus of a point that is always 150 km from station A. The scale gives 150 km = 3 cm, so draw an arc with a radius of 3cm, with A as the center.



Construct the reception area for radio station B. This time draw an arc with the compass set to 3cm, with B as the center. The interference occurs in the area where the two paths overlap.